



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Computer Science & Engineering
End Semester Examination
Academic Year: 2025-26 (Semester – I)

Class: TY B. Tech.	Subject: Elective: Introduction to Machine Learning	Subject Code: PEC-CS307
Date: 21/11/2025	Time: 10:00-12:00	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.

Q.N.	Solve all Questions	Marks	CO	BT																				
Q.1	Explain the different stages of the Machine Learning Life Cycle.	10	CO1	BT2																				
Q.2A)	Given Dataset:<table><tr><th>Point</th><th>X</th><th>Y</th><th>Class</th></tr><tr><td>A</td><td>1</td><td>2</td><td>Red</td></tr><tr><td>B</td><td>2</td><td>3</td><td>Red</td></tr><tr><td>C</td><td>4</td><td>4</td><td>Blue</td></tr><tr><td>D</td><td>6</td><td>5</td><td>Blue</td></tr></table> Classify the point P(3,2) using Euclidean distance. Show all distance calculations and predict new datapoint belongs to which class.	Point	X	Y	Class	A	1	2	Red	B	2	3	Red	C	4	4	Blue	D	6	5	Blue	5	CO2	BT4
Point	X	Y	Class																					
A	1	2	Red																					
B	2	3	Red																					
C	4	4	Blue																					
D	6	5	Blue																					
Q.2B)	Explain the concept of Support Vector Machine (SVM). Describe the hyperplane, support vectors, and margin with the help of a neat labelled diagram.	5	CO2	BT2																				
Q.3A)	Draw and explain Decision Tree with diagram. Mention: root node, internal nodes, branches, leaf nodes, decision rules.	5	CO3	BT2																				
Q.3B)	Differentiate between Supervised and Unsupervised Machine Learning. Explain with suitable examples.	5	CO4	BT2																				
Q.4A)	What is Principal Component Analysis (PCA)? Explain its role in dimensionality reduction.	5	CO5	BT1																				
Q.4B)	Explain Naïve Bayes Classifier with an Example.	5	CO5	BT2																				
Q.5	Explain the architecture of an Artificial Neural Network (ANN) in detail. Describe the role of input layer, hidden layers and output layer with a neat diagram.	10	CO4	BT2																				
Q.6	Given Data points: P1(185,72), P2(170,56), P3(168,60), P4(179,68), P5(182,72), P6(188,77). Initial centroids are: C1 = P1 (185,72), C2 = P2 = (170,56). Perform one full iteration of K-means clustering (Euclidean distance) on the above data points with k=2. Show calculations for distances, cluster assignment, and new centroids.	10	CO2	BT3																				

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand the basic concepts, strengths and weaknesses of many popular machine learning approaches.
CO 2	Learn the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
CO 3	Apply popular machine learning techniques and computing environment that are suitable for the applications under consideration.
CO 4	Identify and Analyze various ways of selecting suitable model parameters for different machine learning techniques.
CO 5	Design and implement various machine learning algorithms in a range of real-world applications.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-2026

Semester: I

Examination: END TERM

Class: T.Y. (CSB)

Branch: CSE

Date: 21/11/2025
Time: 10:00 To 12:00

Subject: Design and Analysis of Algorithms

Code: PCC-CS305

Max. Marks: 60

- Attempt Any FIVE of the following questions. Figures to right indicates Full Marks.
- Illustrate your answers with neat diagrams and examples wherever necessary.

		Marks	CO NO	BT LEVEL
Q.1.	Define Asymptotic Notations: Big-O (O), Big-Omega (Ω), and Big-Theta (Θ). Explain their significance in algorithm analysis with a simple example for each. Also Explain the Master Theorem and Solve the following recurrence relation using it. State which case of the theorem applies and why. $T(n) = 9T(n/3) + n^2$	(12)	CO1	3
Q.2.	Write the algorithm for Quick Sort. Explain the "partition" step clearly. Also, Analyze the best-case, average-case, and worst-case time complexity of Quick Sort. How can the worst-case scenario be practically avoided?	(12)	CO2	4
Q.3.	Write and explain Dijkstra's algorithm for finding the single-source shortest path. Also, Find the shortest path from source vertex 'A' to all other vertices in the following graph using Dijkstra's algorithm. Show the steps clearly in a table format.	(12)	CO4	5
Q.4.	Explain the Matrix Chain Multiplication (MCM) problem. Write the general recurrence relation $m[i, j]$ used to solve it using dynamic programming. Clearly define all variables used in the recurrence (e.g., i, j, k , and the dimension array p). Also write the Algorithm using Dynamic Programming to solve it. You are given a chain of four matrices $\{A_1, A_2, A_3, A_4\}$ with a dimension array $p = [30, 35, 15, 5, 10]$. Find the minimum number of scalar multiplications needed to compute this product. Show your calculations and fill the cost table m (for minimum costs) and the split (Crack) table s (for optimal split points). Write the parenthesized output by referring s table	(12)	CO6	5
Q.5.	Explain the primary advantage of the Bellman-Ford algorithm over Dijkstra's algorithm for finding single-source shortest paths. In your explanation, describe the specific condition under which Bellman-Ford can report a failure and how it detects this condition. Consider the following weighted, directed graph with source vertex A.	(12)	CO4	5
	First draw the given graph and Run the Bellman-Ford algorithm on this graph. - Show the distance $d[v]$ for each vertex v after each pass (i.e., after $k=1, k=2, \dots, k= V -1$ passes). Present this as a table. Here V indicates total number of vertices in the graph. - After all passes are complete, state the final shortest-path distances.			
Q.6.	A) Clearly differentiate between the P, NP, and NP-Complete complexity classes. Explain the relationship between them, and what it would mean if it were proven that $P = NP$. (6 Marks) B) Write Notes on: Minimum Cost Spanning Tree and Prim's Algorithm. (6 Marks)	(12)	CO5 CO4	4

NOTE: - CO: Course Outcomes, BT: Bloom's Taxonomy



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Computer Science & Engineering
End Semester Examination
Academic Year: 2025-26 (Semester – I)

Class: TY B. Tech.	Subject: Operating Systems	Subject Code: PCC-CS303/304
Date: 17/11/2025	Time: 10:00-12:00	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.

Q.N.	Solve all Questions:	Marks	CO	BT
Q.1	Draw and Explain 9 State Transition Diagram of the Process.	10	CO1	BT2
Q.2 A)	Write a simple c program to create pthreads, Explain the arguments passed while creating a thread.	5	CO2	BT4
Q.2 B)	What are different types of threads? Explain with neat diagram	5	CO2	BT2
Q.3	Explain in brief the meaning of Deadlock. What are the conditions for deadlock to occur? What should be done to prevent deadlock?	10	CO2	BT3
Q.4 A)	Explain the working of the Buddy and Slab Allocator with a neat diagram.	5	CO3	BT2
Q.4 B)	Explain the following terms: i) live locks ii) convoy effect iii) Thundering herd Problem.	5	CO4	BT1
Q.5	Explain different Multiprocessor Scheduling techniques in detail.	10	CO5	BT2
Q.6	Explain the functions of memory subsystem. Also explain the actual functions of memory subsystem.	10	CO6	BT3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand Operating System Structure, Operations and Services.
CO 2	Learn the Process Concept, Multithreaded Programming, Process Scheduling and Synchronization.
CO 3	Apply the Concepts of Virtual Memory Management and File Systems.
CO 4	Analyze the Secondary Storage and I/O Systems.
CO 5	Design and implement CPU Scheduling algorithms, Page Replacement Algorithms, Memory Allocation Algorithms, Disk Scheduling Algorithms.
CO6	Understand Memory Management Concepts.

	Shri Guru Gobind Singhji Institute of Engineering & Technology, Vishnupuri, Nanded-431606. Department of Computer Science and Engineering End-Semester Examination Academic Year: 2025-26 (semester I)		
Class: Third Year B.Tech	Course: Formal Language and Automata Theory	Course Code: PCC-CS301/ PCC-CS303	
Date: 19-11-2025, Wed	Time: 10:00 to 12:00	Max. Marks: 60	

Note: Assume suitable data if required.

Q.N.	All Questions are compulsory	CO	BT
Q.1	Attempt the following questions. (Each carries 2 marks)		
a.	The number of states in the minimum sized DFA that accepts the language defined by the regular expression $(0+1)^*(0+1)(0+1)^*$ is _____. a.1 b. 2 c. 3 d.4	CO1	BT1
b.	Which one of the following regular expressions is NOT equivalent to the regular expression $(a+b+c)^*$? a. $(a^*+b^*+c^*)^*$ b. $(a^*b^*c^*)^*$ c. $((ab)^* + c^*)^*$ d. $(a^*b^* + c^*)^*$	CO2	BT1
c.	In the context –free grammar below, S is the start symbol, a and b are terminals, and ϵ denotes the empty string. $S \rightarrow aSAb / \epsilon$ $A \rightarrow bA / \epsilon$ The grammar generates the language a. $((a+b)^*b)^*$ b. $\{a^m b^n \mid m \leq n\}$ c. $\{a^m b^n \mid m = n\}$ d. a^*b^*	CO3	BT1
d.	Consider the following mapping and find the correct language? $\delta(q_0, a, z_0) = (q_1, z_0)$ $\delta(q_0, b, z_0) = (q_2, z_0)$ $\delta(q_1, a, z_0) = (q_1, z_0)$ $\delta(q_1, b, z_0) = (q_2, z_0)$ $\delta(q_2, a, z_0) = (q_2, z_0)$ $\delta(q_2, b, z_0) = (q_0, z_0)$ $\delta(q_1, \epsilon, z_0) = (q_f, z_0)$ a. ending with a b. ending with a, contain even number of a c. ending with a, contain odd number of a d. ending with a, contain even number of b	CO4	BT1
e.	Which of the following statements is TRUE? a. Every recursive language is recursively enumerable b. Every recursively enumerable language is recursive c. Every regular language is context-free but not recursive d. Every recursive language is regular	CO5	BT1
Q.2	Attempt the following questions.(Each carries 4 marks)		
a.	Identify the language generated by the given regular expressions 1. $(a+b)(a+b)^+$ 2. $(a+b)(a+b)$	CO2	BT3
b.	Convert the given CFG into its equivalent CNF. $S \rightarrow a / aA / B$ $A \rightarrow aBB / \epsilon$ $B \rightarrow Aa/b$	CO2	BT3
c.	Prove that $L = \{a^p \mid p \text{ is prime}\}$ is not regular.	CO2	BT4

d.	Convert the following NFA shown in Figure 1 to its equivalent DFA. Figure 1	CO2	BT3
e.	Find regular expressions for the following FA using the state elimination method.	CO2	BT1
Q.3	Attempt the following questions. (Each carries 6 marks)		
a.	Given the grammar. $S \rightarrow aSb \mid ab$ Construct a PDA that accepts the same language generated by this grammar.	CO4	BT6
b.	Convert the following ϵ-NFA into an equivalent DFA by using ϵ-closure method.	CO2	BT3
c.	Explain the limitations of: 1. Finite Automata (FA) 2. Pushdown Automata (PDA) 3. Turing Machine (TM)	CO6	BT4
d.	Explain the Chomsky hierarchy with a suitable examples.	CO4	BT4
e.	Construct a Turing Machine that computes the 2's complement of a binary number. Give the instantaneous description for the input string "1001010" by using the resultant Turing machine.	CO5	BT6

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

CO's	Statement
CO1	Understand the fundamental concepts of automata and formal languages.
CO2	Analyze and construct finite automata (FA) and regular expressions (RE).
CO3	Demonstrate knowledge of context-free grammars (CFGs) and context-free languages (CFLs).
CO4	Examine pushdown automata (PDA) and their relationship with context-free grammars.
CO5	Understand how Turing machines (TM) and computational theory work
CO6	Explore the applications and limitations of formal language theory.

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create



Class: Third Year

Course Name: MDM-
Introduction to Cryptography

Course Code: MDM-IS-
CS301

Date: 28/11/2025

Time: 10.00 to 12.00

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	A	I. Perform addition: $(x^3 + x + 1) + (x^2 + x)$ over $GF(2)$.	3	CO1	BT3
		II. Give multiplication table for Z_5 .	2	CO1	BT3
	B	I. Find whether 97 is prime using the square root method.	3	CO1	BT3
		II. Define modulo multiplicative inverse.	2	CO1	BT3
Q.2	A	Alice and Bob, agree to use a prime modulus $p=23$ and a primitive root $g=5$. Alice chooses a private key $a=6$. Bob chooses a private key $b=15$. a) Compute Alice and Bob's public key A and B respectively. c) Compute the shared secret key for both Alice and Bob.	5	CO2	BT6
	B	Evaluate different authentication mechanisms (password, token, biometric) and their practical trade-offs.	5	CO4	BT5
Q.3	A	Compare Discretionary, Mandatory, and Role-Based Access Control models with appropriate examples.	5	CO5	BT4
	B	Evaluate how digital signatures enhance integrity and non-repudiation in modern communication systems.	5	CO2	BT5
Q.4	A	Alice wish to send signed message to Bob. He selects $p = 7$, $q = 17$ $e = 5$ and message $M = 9$. Give complete scheme of RSA digital signature with a diagram.	5	CO2	BT6
	B	Analyse cipher feedback and output feedback modes. Give their typical applications	5	CO2	BT4
Q.5	A	Attempt any one I. Write note on Message Authentication Codes II. Write note on Hash Functions	5	CO3	BT2
	B	Attempt any one I. Explain how firewalls enforce organizational network security policies through packet filtering and rule sets. II. Explain how VPNs use tunneling and encryption to ensure confidentiality and integrity in communications.	5	CO6	BT2

Q6.		Attempt any two			
	a.	Explain in brief Double DES and Tripple DES.	5	CO2	BT2
	b	Examine why Double DES does not effectively double the key strength as expected.	5	CO2	BT4
	c	Use the Playfair cipher with key SECURITY to encrypt plaintext: BALLOON ARRIVES TODAY.	5	CO2	BT2
	d	Define Confidentiality, Integrity, and Availability with suitable examples.	5	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

CO1: Understand the number theory concepts.

CO2: Impart knowledge on symmetric and asymmetric encryption techniques.

CO3: Create Authentication functions as Message Authentication Codes and Hash Functions

CO4: Examine the issues and structure of Authentication Service and Electronic Mail, web and IP security.

CO5: Provide familiarity in system security.

CO6: Understand and apply the principles of cryptographic protocols and network security to ensure secure communication and data protection.